

Global Valuation

Cross-Country Valuation Multiples: Data and First Results

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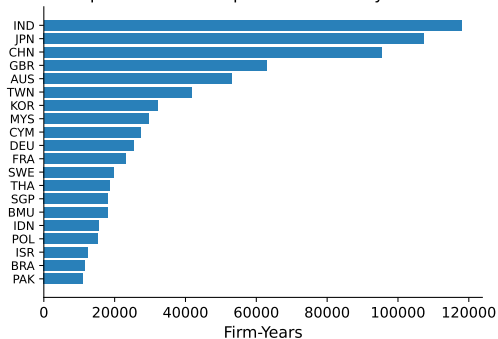
Datasets

File	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (124 countries)
compustat_america	0.8 GB	254K	984	NA fundamentals + CRSP link
returns_global	1.9 GB	10.4M	25	Global monthly prices
returns_america	0.9 GB	3.6M	55	CRSP US monthly returns

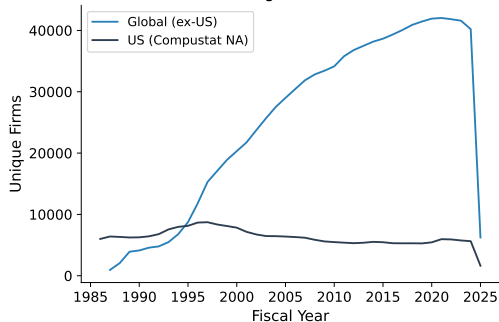
- Compustat Global: fiscal years 1987–2024, 124 countries
- Compustat America: fiscal years 1986–2024, CRSP-linked
- Combined regression panel: **743,957 firm-years**, 72,787 firms, 81 countries

Coverage: Countries and Time

Compustat Global: Top 20 Countries by Firm-Years



Coverage Over Time



India, Japan, China dominate non-US coverage. Panel peaks at ~41K firms (2022).

Panel Construction

- **Market-to-book:** $\log(M/B) = \log(\text{Market Cap}/\text{Book Equity})$
 - US: `mkvalt` from Compustat NA; Global: `cshoi` $\times$ `prccm`
- **ROA:** Net Income / Total Assets **Size:** $\log(\text{Total Assets})$
- **Lag earnings growth:** $(E_{t-1} - E_{t-2})/|E_{t-2}|$ **Industry:** GICS Ind. Group
- Winsorize M/B and ROA at 1st/99th percentiles; filter country-years with ≥ 30 obs

Variable	N	Mean	P10	Median	P90
$\log(M/B)$	743,957	0.46	-0.77	0.39	1.76
ROA	743,523	-0.01	-0.15	0.02	0.11
$\log(\text{Assets})$	743,957	7.28	3.26	7.17	11.49

Cross-Country Regressions

Regression Specifications

Market-to-Book:

$$\begin{aligned}\log(M/B)_{i,t} = & \alpha + \beta_1 \text{ROA}_{i,t} + \beta_2 \log(\text{Assets})_{i,t} + \beta_3 \Delta E_{i,t-1} \\ & + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}\end{aligned}\quad (\text{Spec 1})$$

$$\begin{aligned}\log(M/B)_{i,t} = & \alpha + \beta_1 \text{ROA}_{i,t} + \beta_2 \log(\text{Assets})_{i,t} + \beta_3 \Delta E_{i,t-1} \\ & + \delta_j + \gamma_{c,t} + \varepsilon_{i,t}\end{aligned}\quad (\text{Spec 2})$$

ROA:

$$\text{ROA}_{i,t} = \alpha + \beta_1 \log(\text{Assets})_{i,t} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

γ_c = country FE, δ_j = GICS industry group FE, τ_t = year FE, $\gamma_{c,t}$ = country $\times$ year FE.
Standard errors clustered at the country level.

M/B Regression Results

	Spec 1	Spec 2
	C + Ind + Year FE	C×Year + Ind FE
ROA	0.015 (0.136)	−0.020 (0.133)
log(Assets)	−0.031 (0.025)	−0.033 (0.025)
Lag Earn. Growth	0.006*** (0.002)	0.005** (0.002)
N / R^2	593,530 / 0.180	593,530 / 0.206
C / Ind / Year FE	Y / Y / Y	− / Y / −
C × Year FE	−	Yes

*** $p < 0.01$. Clustered SEs at country level.

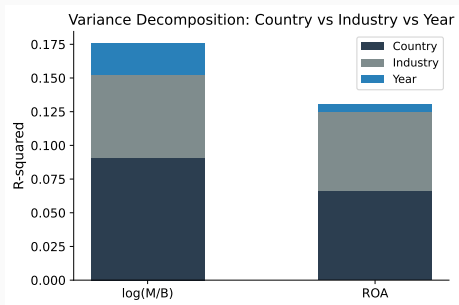
Key Takeaway: ROA Does Not Explain M/B

- ROA coefficient is **insignificant** in both specifications
- After country, industry, and year controls, current profitability does not predict M/B
- Only **lag earnings growth** is significant (positive, as expected)
- Size effect negative but insignificant with clustered SEs
- **Interpretation:** cross-country M/B differences reflect *discount rates* and *growth expectations*, not current profitability

R^2 **comparison:** Spec 1 = 0.180, Spec 2 = 0.206 — country×year interactions gain 2.6 pp, suggesting time-varying country effects matter.

Variance Decomposition

What Drives Valuation? Country vs Industry vs Year



Shapley R^2 decomposition: log(M/B):

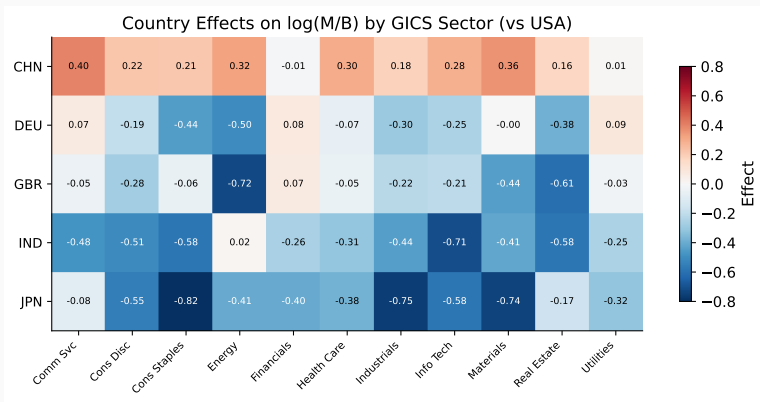
- Country: **51.6%** of R^2
- Industry: 35.1%
- Year: 13.3%

ROA:

- Country: **50.9%** of R^2
- Industry: 44.7%
- Year: 4.4%

Country dominates both valuation and profitability.

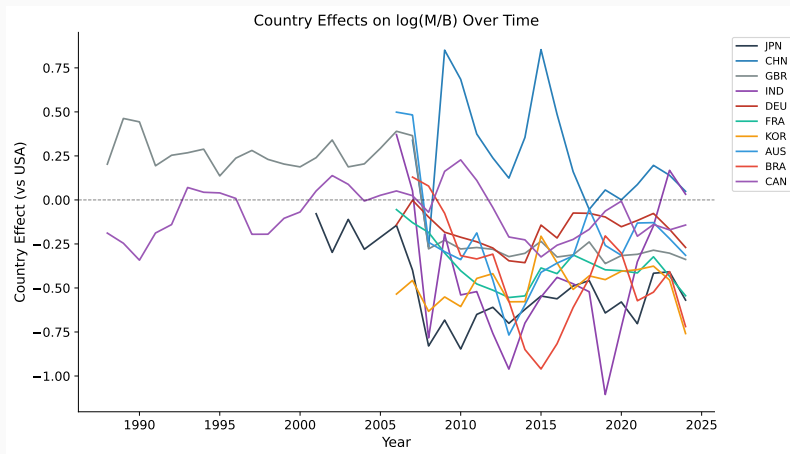
Industry × Country Interaction



Country effects on log(M/B) by GICS sector (vs USA). Adding Country×Sector interaction raises R^2 by +2.8 pp — country effects vary meaningfully across industries.

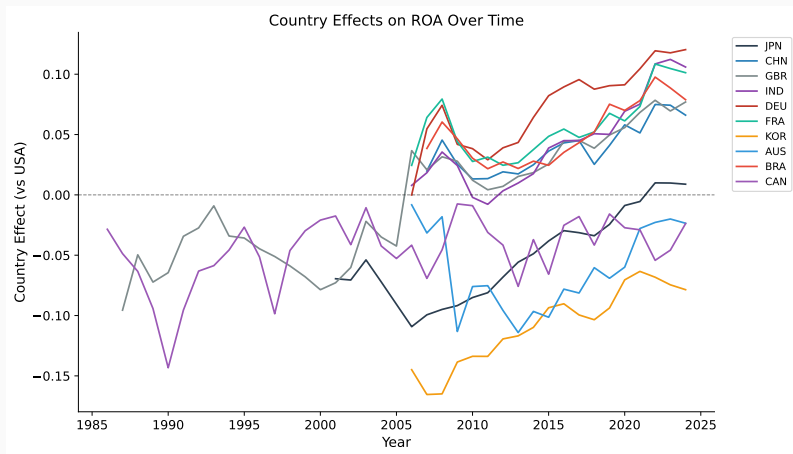
Country Effects Over Time

Country Effects on M/B (vs USA)



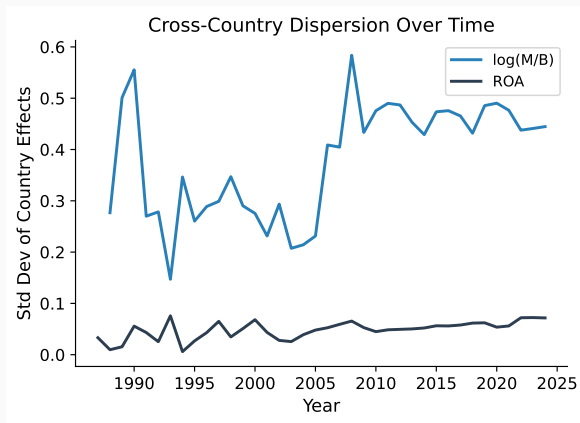
Year-by-year: $\log(M/B) \sim \text{ROA} + \log(\text{Assets}) + \text{Country FE} + \text{Industry FE}$. USA = 0.

Country Effects on ROA (vs USA)



Year-by-year: $ROA \sim \log(\text{Assets}) + \text{Country FE} + \text{Industry FE}$. USA = 0.

Cross-Country Dispersion Over Time



No convergence in valuations:

- M/B dispersion: $0.40 \rightarrow 0.47$
- Trend is **diverging**
- Spike during 2008–09 GFC

ROA dispersion more stable:

- Lower level (~ 0.05 – 0.10)
- Spike in 2009–10 (earnings collapse)

Valuation gaps are *widening* despite globalization.

Stylized Facts

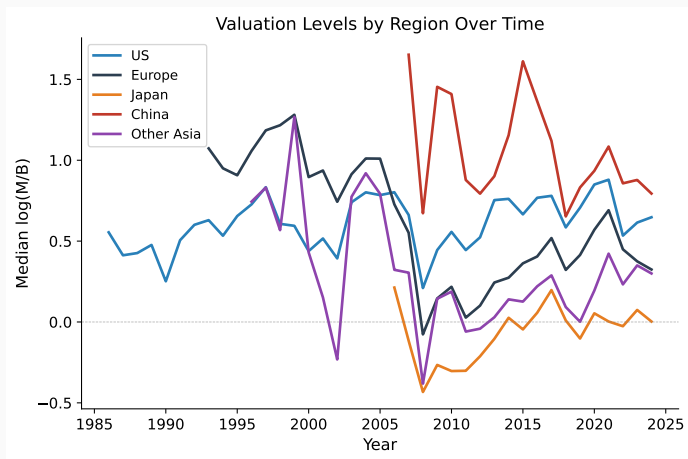
Stylized Facts (1/2): Decomposition

1. **Country effects dominate valuation** — country explains 52% of M/B R^2 (Shapley), vs 35% for GICS industry. Echoes the Heston–Rouwenhorst debate, but for *levels* rather than returns.
2. **Country dominates profitability too** — for ROA, country edges industry (51% vs 45%). Both valuation and profitability are primarily location-driven.
3. **Negative M/B–ROA correlation across countries** ($\rho = -0.26$) — high-M/B countries are *not* high-ROA countries. Discount rates and growth expectations, not current profitability, drive the cross-country valuation map.

4. **Japan discount is persistent** — M/B effect negative in **100%** of years (avg -0.54 vs USA), with no sign of convergence.
5. **China premium collapsed post-2015** — from $+0.8$ to near zero, consistent with market correction and deleveraging.
6. **Cross-country valuation dispersion is increasing** — despite financial globalization, countries are *diverging* in M/B (std dev: $0.40 \rightarrow 0.47$).

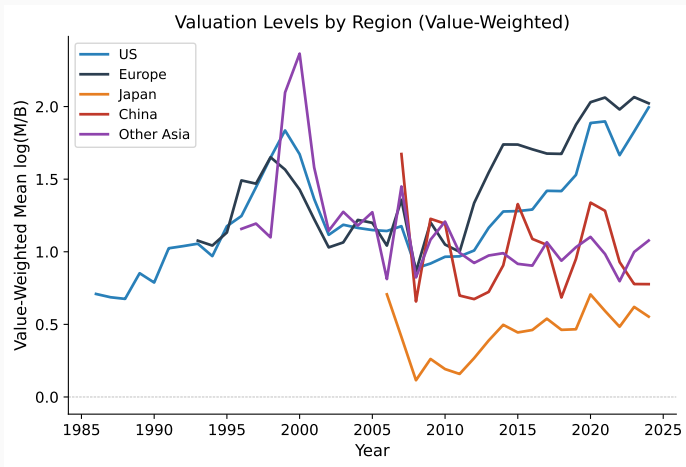
Regional Trends

Regional Valuations: Equal-Weighted Median



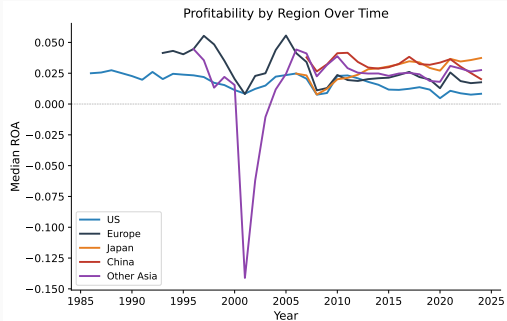
Median log(M/B) by region-year. China separately shown; Japan persistently lowest.

Regional Valuations: Value-Weighted

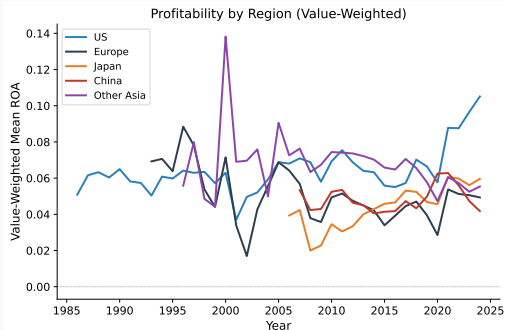


VW mean $\log(M/B)$: cap-weighted within country, then EW across countries.

Regional Profitability Over Time



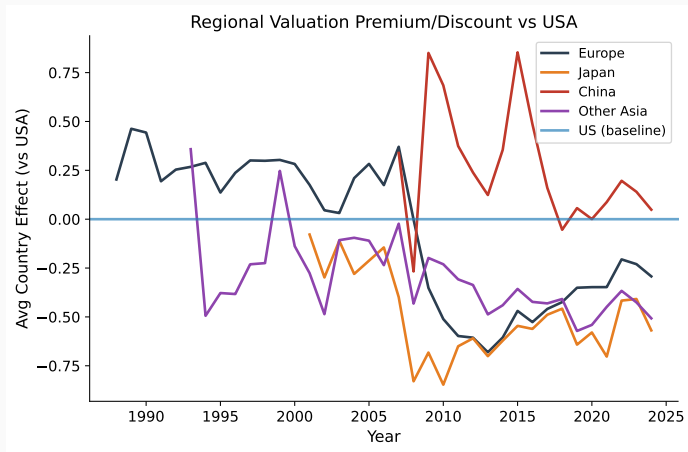
EW median



Value-weighted

Japan and Other Asia have *higher* profitability than US despite lower valuations.

Regional Valuation Gap vs USA



Average country effect (vs USA baseline) from year-by-year regressions with GICS industry FE.

The US Valuation Premium

- **EW median gap** (recent 5 years): US +25% vs Europe, +98% vs Japan, +49% vs Other Asia
- **China** has comparable M/B to US on EW median — but declining since 2015
- Japan and Other Asia have *higher median ROA* than the US — yet *lower valuations*
- VW results show even larger gaps: US +258% vs Japan, +129% vs Other Asia

Central question: Is the US premium driven by higher *growth expectations* or lower *discount rates*?

Growth vs Discount Rates

Two Hypotheses: Growth vs Discount Rates

Gordon growth model: $M/B \propto \frac{ROE-g}{r-g}$

Hypothesis 1: Growth expectations (g)

- High-M/B countries have higher expected earnings growth
- Bekaert et al. (2007, *JF*): liberalization raises growth opportunities
- Pastor–Veronesi (2003, *JF*): profitability uncertainty raises M/B

Hypothesis 2: Discount rates (r)

- Low-M/B countries have higher required returns
- Hail–Leuz (2006, *JAR*): institutions reduce cost of equity by up to 200 bps
- Stulz (2005, *JF*): expropriation risk keeps discount rates high
- Bekaert–Harvey (2000, *JF*): segmentation raises required returns

Evidence Against Pure Growth Story

- **Asia's ROA is higher than the US** — but valuations are lower. If growth tracked profitability, Asia should have *higher* M/B.
- **Negative M/B–ROA correlation** ($\rho = -0.26$): high-profitability countries are *not* high-valuation countries.
- **Japan**: strong ROA, yet deepest discount (-0.55 vs USA).
- Claus–Thomas (2001, *JF*): implied equity premia vary modestly (3–5%) across G-7.

⇒ Current profitability does not explain the valuation map. **Discount rates and future growth expectations** are the key drivers.

Why discount rates differ across countries:

- **Investor protection:** LLSV (2002) — common-law Q premium of ~ 0.25 . Djankov et al. (2008) — anti-self-dealing index predicts valuations.
- **Market segmentation:** Bekaert–Harvey (1995, 2000) — segmented markets price local risk. Bekaert et al. (2011) — segmentation persists.
- **Cross-listing:** Doidge et al. (2004) — 17% Q premium from US governance bonding. Hail–Leuz (2009) — cost of capital falls after listing.
- **Sovereign risk:** Erb et al. (1996) — credit ratings predict expected returns.

The investor-side literature is deeper than the growth-side literature.

Literature

Related Literature: Decomposition and Institutions

Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country effects dominate for returns. We find the same for *valuation levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects matter more in traded-goods sectors.
- Brooks–Del Negro (2004) — country effects declined post-1999 for returns. Opposite for M/B.

Legal origin and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q.
- Djankov et al. (2008, *JFE*) — anti-self-dealing index predicts valuations.
- Doidge et al. (2007, *JFE*) — countries matter more than firms for governance.

Related Literature: Globalization and Dynamics

Limits of convergence:

- Stulz (2005, *JF*) — “twin agency problems” prevent equalization.
- Bekaert et al. (2011, *RFS*) — segmentation decreased but persists.
- Doidge et al. (2004, *JFE*) — cross-listed firms gain $\sim 17\%$ Q premium.

Country-specific puzzles and fundamentals:

- French–Poterba (1991, *JFE*) — Japan P/E premium; persistent discount since.
- Fama–French (1998, *JF*) — value premium exists internationally.
- Rajan–Zingales (1998, *AER*) — financial development raises valuations.

Our Contribution

- First panel decomposition of *valuation levels* (not returns) into country, industry, and time using Shapley R^2
- Largest cross-country valuation panel: 81 countries, 1986–2024, 744K firm-years
- Document **divergence** in M/B across countries, challenging the globalization-implies-equalization view
- Country M/B effects **not** explained by ROA ($\rho = -0.26$) — points to discount-rate and institutional channels